3-Phase Grid-Tied Inverter

Internship Week 1 Progress

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Project Timeline & Initial Testing

- Generated simple PWM in C2000 MCU.
- Implemented complimentary PWM with dead band.
- Tested on Tolarix half-bridge module (1kΩ resistance, 15V supply).
- Developed sinusoidal PWM generation.
- Extended to 3-phase Sine PWM with 120° phase difference.
- **Verified all above waveforms on oscilloscope.**
- Transitioned to understanding Phase-Locked Loops (PLL).
- Conducted PLL and Grid-Tied simulations in MATLAB.

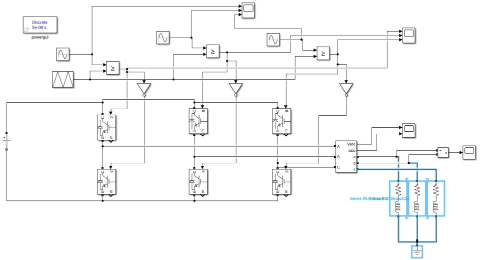
[1]

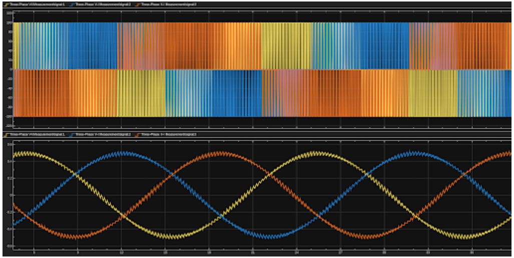
3-Phase Sine PWM Code (C2000)

```c
#include "F28x_Project.h"
#include <math.h>
#define PI 3.14159265
#define PWM_FREQ 5000.0
#define FUND_FREQ 50.0

volatile Uint16 DeadBand = 500;
volatile float ModIndex = 0.8;
volatile float theta = 0.0;
volatile float delta_theta;

void InitEPwm1(void);


void main(void) {
    InitSysCtrl();
    CpuSysRegs.PCLKCR2.bit.EPWM1 = 1; CpuSysRegs.PCLKCR2.
          bit.EPWM2 = 1; CpuSysRegs.PCLKCR2.bit.EPWM3 = 1;
    InitEPwm1Gpio(); InitEPwm2Gpio(); InitEPwm3Gpio();
    DINT; InitPieCtrl(); IER = 0; IFR = 0; InitPieVectTable
          ();
    EALLOW; CpuSysRegs.PCLKCR0.bit.TBCLKSYNC = 0; EDIS;
    InitEPwm1(); /* InitEPwm2(); InitEPwm3(); */
    EALLOW; CpuSysRegs.PCLKCR0.bit.TBCLKSYNC = 1; EDIS;

    delta_theta = 2.0 * PI * FUND_FREQ / PWM_FREQ;

    while(1) {
        EPwm1Regs.DBRED.bit.DBRED = DeadBand; EPwm1Regs.
              DBFED.bit.DBFED = DeadBand;
        if(ModIndex > 1.0) ModIndex = 1.0;
        if(ModIndex < 0.0) ModIndex = 0.0;

        EPwm1Regs.CMPA.bit.CMPA = (Uint16)((EPwm1Regs.TBPRD
              /2.0)*(1.0+ModIndex*sin(theta)));
        EPwm2Regs.CMPA.bit.CMPA = (Uint16)((EPwm2Regs.TBPRD
              /2.0)*(1.0+ModIndex*sin(theta+2*PI/3)));
        EPwm3Regs.CMPA.bit.CMPA = (Uint16)((EPwm3Regs.TBPRD
              /2.0)*(1.0+ModIndex*sin(theta+4*PI/3)));

        theta += delta_theta;
        if(theta >= 2.0*PI) theta -= 2.0*PI;
        DELAY_US(200);
    }
}
void InitEPwm1(void) {
    EPwm1Regs.TBPRD = 20000; EPwm1Regs.TBPHS.bit.TBPHS = 0;
           EPwm1Regs.TBCTR = 0;
    EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UPDOWN;
          EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE;
    EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1; EPwm1Regs.
          TBCTL.bit.CLKDIV = TB_DIV1;
    EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW; EPwm1Regs.
          CMPCTL.bit.LOADAMODE = CC_CTR_ZERO;
    EPwm1Regs.CMPA.bit.CMPA = 10000;
    EPwm1Regs.AQCTLA.bit.CAU = AQ_CLEAR; EPwm1Regs.AQCTLA.
          bit.CAD = AQ_SET;
    EPwm1Regs.DBCTL.bit.OUT_MODE = DB_FULL_ENABLE;
          EPwm1Regs.DBCTL.bit.POLSEL = DB_ACTV_HIC;
    EPwm1Regs.DBCTL.bit.IN_MODE = DBA_ALL;
    EPwm1Regs.DBRED.bit.DBRED = DeadBand; EPwm1Regs.DBFED.
          bit.DBFED = DeadBand;
}
```

Sine PWM Simulation & Waveforms

- Sinusoidal PWM with dead band and complementary signals.
- Open loop 3-phase simulation.



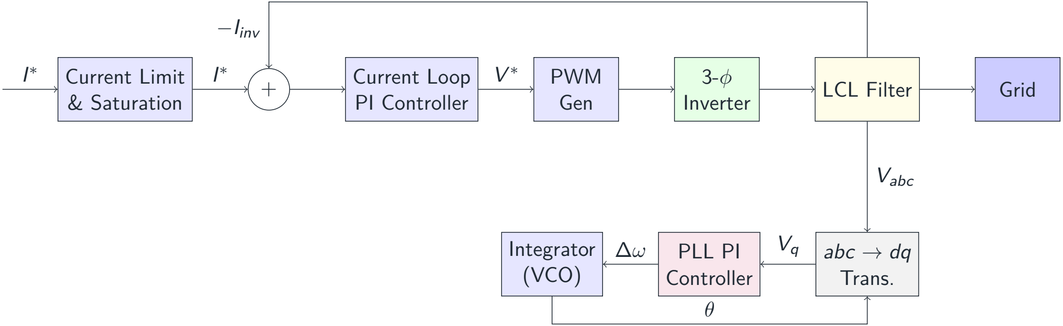


Theory: Phase-Locked Loops (PLL)

- **Challenge:** Grid currents are AC. PI controllers struggle to track AC without error.
- **Solution (abc to dq):** Transform 3-phase AC into a synchronously rotating frame (dq). AC variables appear as DC.
- d-axis controls **active power**, q-axis controls **reactive power**.
- **Role of PLL:** Estimates grid angle θ required for the Park Transform by driving V_q to zero.

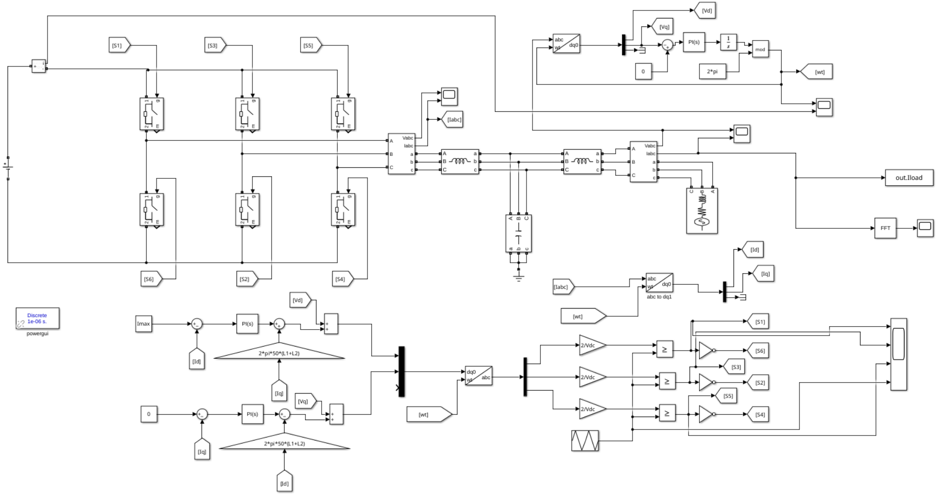
[2], [3]

Overall Block Diagram



[4]

PLL Block Diagram (Simulink)



[4]

System Specifications

- **Grid Voltage ($V_{g,LL}$):** 400 V (Line-to-Line RMS)
- **Grid Frequency (f_g):** 50 Hz
- **Rated Power (P_{rated}):** 10 kW
- **DC Link Voltage (V_{dc}):** 700 V
- **Switching Frequency (f_{sw}):** 10 kHz

Derived Base Values

Phase voltage: $V_{ph,RMS} = 400/\sqrt{3} = 230.94$ V
Peak current: $I_{ph,peak} = \sqrt{2} \cdot 10000/(3 \cdot 230.94) = 20.41$ A

[5]

LCL Filter Design (1/2)

Inverter-Side Inductor (L_1)

Ripple current limit: $\leq 15\%$ of $I_{ph,peak} = 3.06$ A

$$L_1 \geq \frac{V_{dc}}{8 \cdot f_{sw} \cdot \Delta I} = \frac{700}{8 \cdot 10000 \cdot 3.06} = 2.86 \text{ mH} \implies \textbf{Choose 3.0 mH}$$

Filter Capacitor (C_f)

Reactive power limit: $\leq 5\%$ of P_{rated}

$$C_f \leq \frac{0.05 \cdot P_{rated}}{2\pi f_g \cdot V_{ph,RMS}^2} = \frac{500}{2\pi \cdot 50 \cdot (230.94)^2} = 29.8\ \mu\text{F} \implies \textbf{Choose 20 }\boldsymbol{\mu}\textbf{F}$$

[6]

LCL Filter Design (2/2)

Grid-Side Inductor (L_2)

$$L_2 = 0.5 \cdot L_1 = 0.5 \cdot 3.0 \text{ mH} = \mathbf{1.5 \text{ mH}}$$

Resonance Frequency (f_{res}) & Damping Resistor (R_d)

Must satisfy $10f_g < f_{res} < 0.5f_{sw}$ (500 Hz $< f_{res} <$ 5000 Hz)

$$f_{res} = \frac{1}{2\pi}\sqrt{\frac{L_1 + L_2}{L_1 L_2 C_f}} = \frac{1}{2\pi}\sqrt{\frac{4.5\text{m}}{4.5\mu \cdot 20\mu}} \approx \mathbf{796 \text{ Hz}} \quad (\checkmark)$$

$$R_d = \frac{1}{3 \cdot 2\pi f_{res} C_f} = \frac{1}{3 \cdot 2\pi \cdot 796 \cdot 20\mu} \approx \mathbf{3.3}\ \Omega$$

[6]

Current Loop PI Tuning

- Total inductance $L_{tot} = 4.5$ mH, parasitic resistance $R_{tot} \approx 0.1\ \Omega$.
- Bandwidth $\alpha = 2\pi \cdot (f_{sw}/10) = 3142$ rad/s.

PI Gains Calculation

$$K_p = \alpha \cdot L_{tot} = 3142 \cdot 0.0045 = \mathbf{14.14\ V/A}$$

$$K_i = \alpha \cdot R_{tot} = 3142 \cdot 0.1 = \mathbf{314.2\ V/(A \cdot s)}$$

Decoupling Terms ($v_{d,inv}$ and $v_{q,inv}$)

Cross-coupling removal: $\omega L_{tot} = 2\pi(50)(0.0045) = 1.414\ \Omega$

$$v_{d,inv} = \mathrm{PI}_d - \omega L_{tot} i_q + v_{g,d}, \qquad v_{q,inv} = \mathrm{PI}_q + \omega L_{tot} i_d$$

[3]

PLL PI Tuning

- Choose natural frequency $\omega_n = 30$ rad/s and damping $\zeta = 0.707$.

PI Gains Calculation

$$K_{p,pll} = 2\zeta\omega_n = 2(0.707)(30) = \mathbf{42.42}$$

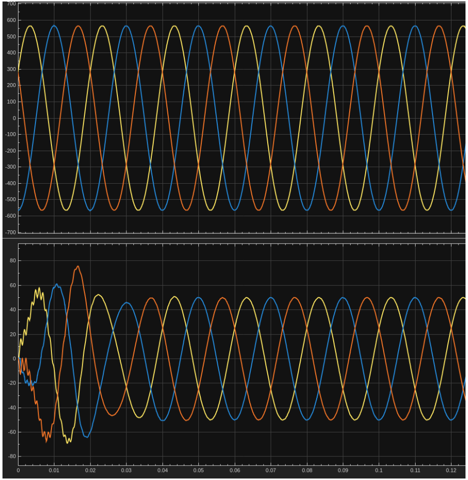
$$K_{i,pll} = \omega_n^2 = 30^2 = \mathbf{900}$$

PLL Output Integration

$$\omega_{pll} = K_{p,pll} \cdot v_q + K_{i,pll} \int v_q \, dt, \qquad \theta = \int \omega_{pll} \, dt$$

[3]

PLL Simulation Waveforms



[3]

Final Parameter Summary

- Evaluated for 10kW, 400V Grid, 10kHz switching.

Component/Controller	Symbol	Computed Value
Inverter-side inductor	L_1	3.0 mH
Grid-side inductor	L_2	1.5 mH
Filter capacitor	C_f	20 μF
Damping resistor	R_d	3.3 Ω
Current Loop Proportional	K_p	14.14 V/A
Current Loop Integral	K_i	314.2 V/(A·s)
PLL Proportional	$K_{p,pll}$	42.42
PLL Integral	$K_{i,pll}$	900

References I

[1] *Tms320f2837xd dual-core real-time microcontrollers technical reference manual*, Texas Instruments.

[2] "Direct-quadrature-zero transformation," Wikipedia. (), [Online]. Available: https://en.wikipedia.org/wiki/Direct-quadrature-zero_transformation.

[3] L. Umanand, "Nptel course," (), [Online]. Available: https://onlinecourses.nptel.ac.in/e-learning/preview/noc20_ee57.

[4] "Matlab simulink block diagram for pll," (), [Online]. Available: https://www.youtube.com/watch?v=V_wfVstfxMc.

[5] B. Singh, A. Chandra, and K. Al-Haddad, *Power Quality Problems and Mitigation Techniques*. Wiley, 2014.

References II

[6] X. Ruan, X. Wang, D. Pan, D. Yang, W. Li, and C. Bao, *Control Techniques for LCL-Type Grid-Connected Inverters* (CPSS Power Electronics Series). Springer, 2018.